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DARK START POWER SUPPLY CALIBRATION AND OPTIMIZER

Abstract

Bidirectional energy transfer systems are provided for transferring energy between an electrified vehicle and other structures. The bidirectional energy transfer system may utilize dark start reserve power from a dark start energy storage resource for maintaining communications between electric vehicle supply equipment (EVSE) and a combiner box during grid power outage conditions. An output voltage of the dark start energy storage resource may be boosted within a variable voltage regulated boost circuit of the combiner box in order to supply a calibrated output voltage to the EVSE. The calibrated output voltage compensates for voltage drops that can occur across a length of a cable that extends between the EVSE and the combiner box.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to bidirectional energy transfer systems capable of transferring energy between an electrified vehicle and other structures.

BACKGROUND

[0002] Plug-in type electric vehicles include one or more charging interfaces for charging a traction battery pack. Plug-in vehicles are typically charged while parked at a charging station or some other utility power source. Plug-in vehicles can also be used to support household loads during electrical power outages.

SUMMARY

[0003] A bidirectional energy transfer system according to an exemplary aspect of the present disclosure includes, among other things, an electric vehicle supply equipment (EVSE), and a combiner box including a dark start energy storage resource and a variable voltage regulated boost circuit configured to boost an output voltage of the dark start energy storage resource for powering a circuitry of the EVSE during a grid power outage condition.

[0004] In a further non-limiting embodiment of the foregoing bidirectional energy transfer system, the dark start energy storage resource is a battery, a capacitor, or a supercapacitor.

[0005] In a further non-limiting embodiment of either of the foregoing bidirectional energy transfer systems, the variable voltage regulated boost circuit is configured to boost the output voltage to compensate for a voltage drop that can occur across a length of a cable that extends from the EVSE to the combiner box.

[0006] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the cable is a Power over Ethernet (POE) cable.

[0007] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, a control system is programmed to control the variable voltage regulated boost circuit for boosting the output voltage during the grid power outage condition.

[0008] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the control system includes a first control module of the combiner box and a second control module of the EVSE.

[0009] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the second control module is programmed to determine a voltage drop based on a comparison between a measured output voltage received from the variable voltage regulated boost circuit and a calibrated output voltage setting received from the first control module.

[0010] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the first control module is programmed to determine a calibrated output voltage for powering the circuitry based on the voltage drop.

[0011] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the first control module is programmed to command the variable voltage regulated boost circuit to adjust the output voltage to the calibrated output voltage for powering the circuitry.

[0012] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the EVSE includes a dark start measurement circuit configured for measuring the output voltage.

[0013] In a further non-limiting embodiment of any of the foregoing bidirectional energy transfer systems, the EVSE further includes a filter and a switched-mode power supply (SMPS).

[0014] A method according to another exemplary aspect of the present disclosure includes, among other things, boosting an output voltage of a dark start energy storage resource of a combiner box during a grid power outage condition, and powering communications between an electric vehicle supply equipment (EVSE) and an electrified vehicle using the boosted output voltage.

[0015] In a further non-limiting embodiment of the foregoing method, the combiner box includes a variable voltage regulated boost circuit that is configured to boost the output voltage.

[0016] In a further non-limiting embodiment of either of the foregoing methods, boosting the output voltage includes sending a calibrated output voltage setting and a first output voltage from the combiner box to the EVSE.

[0017] In a further non-limiting embodiment of any of the foregoing methods, boosting the output voltage further includes measuring the first output voltage within a dark start measurement circuit of the EVSE to determine a measured output voltage received by the EVSE.

[0018] In a further non-limiting embodiment of any of the foregoing methods, boosting the output voltage includes comparing the measured output voltage to the calibrated output voltage setting, and determining a voltage drop.

[0019] In a further non-limiting embodiment of any of the foregoing methods, boosting the output voltage includes sending the voltage drop to the combiner box.

[0020] In a further non-limiting embodiment of any of the foregoing methods, boosting the output voltage includes adjusting the first output voltage to a second output voltage that is derived from the voltage drop.

[0021] In a further non-limiting embodiment of any of the foregoing methods, the second output voltage is a sum of the voltage drop plus a minimum required voltage output that is necessary for maintaining a proper control pilot operation between the EVSE and the electrified vehicle.

[0022] In a further non-limiting embodiment of any of the foregoing methods, in response to establishing the communications between the EVSE and the electrified vehicle, the method includes transferring power from the electrified vehicle to a structure that is separate from the electrified vehicle during the grid power outage condition.

[0023] The embodiments, examples, and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

[0024] The various features and advantages of this disclosure will become apparent to those skilled in the art from the following detailed description. The drawings that accompany the detailed description can be briefly described as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 schematically illustrates a first configuration of a bidirectional energy transfer system.

[0026] FIG. 2 schematically illustrates a second configuration of the bidirectional energy transfer system of FIG. 1.

[0027] FIG. 3 schematically illustrates additional aspects of the bidirectional energy transfer system of FIGS. 1 and 2.

[0028] FIG. 4 is a flow chart of an exemplary method for controlling a bidirectional energy transfer system during a grid power outage condition.

DETAILED DESCRIPTION

[0029] This disclosure relates to bidirectional energy transfer systems capable of transferring energy between an electrified vehicle and other structures. The bidirectional energy transfer system may utilize dark start reserve power from a dark start energy storage resource for maintaining communications between electric vehicle supply equipment (EVSE) and a combiner box during grid power outage conditions. An output voltage of the dark start energy storage resource may be boosted within a variable voltage regulated boost circuit of the combiner box in order to supply a calibrated output voltage to the EVSE. The calibrated output voltage compensates for voltage drops

that can occur across a length of a cable that extends between the EVSE and the combiner box. These and other features of this disclosure are discussed in greater detail in the following paragraphs of this detailed description.

[0030] FIGS. 1, 2, and 3 schematically illustrate an exemplary bidirectional energy transfer system **10** (hereinafter “the system **10**”) for bidirectionally transferring energy between a vehicle **12** and a structure **14**. The system **10** enables the bidirectional transfer of energy from the vehicle **12** to the structure **14** or vice versa. The structure **14** may be a residential building, a commercial building, a parking garage, a charging station, or any other type of structure that is capable of receiving or transferring energy. In an embodiment, the structure **14** is a residential household that functions as a “home location” of the vehicle **12**.

[0031] Although a specific component relationship is illustrated in the figures of this disclosure, the illustrations are not intended to limit this disclosure. The placement and orientation of the various components of the depicted system are shown schematically and could vary within the scope of this disclosure. In addition, the various figures accompanying this disclosure are not necessarily drawn to scale, and some features may be exaggerated or minimized to emphasize certain details of a particular component, assembly, or system.

[0032] In an embodiment, the vehicle **12** is a plug-in type electric vehicle (e.g., a plug-in hybrid electric vehicle (PHEV) or a battery electric vehicle (BEV)). The vehicle **12** includes a traction battery pack **16** that is part of an electrified powertrain capable of applying a torque from an electric machine (e.g., an electric motor) for driving wheels **18** of the vehicle **12**. The electrified powertrain of the vehicle **12** may electrically propel the set of wheels **18** either with or without the assistance of an internal combustion engine.

[0033] The vehicle **12** of FIGS. 1-2 is schematically illustrated as a car. However, other vehicle configurations are also contemplated. The teachings of this disclosure may be applicable for any type of vehicle as the vehicle **12**. For example, the vehicle **12** could be configured as a car, a pickup truck, a van, a sport utility vehicle (SUV), etc.

[0034] Although shown schematically, the traction battery pack **16** may be configured as a high voltage traction battery pack that includes a plurality of battery arrays **20** (e.g., battery assemblies or groupings of battery cells) capable of outputting electrical power to one or more electric machines of the vehicle **12**. Other types of energy storage devices and/or output devices may also be used to electrically power the vehicle **12**.

[0035] The vehicle **12** may interface with the structure **14** through electric vehicle supply equipment (EVSE) **22** in order to perform the bidirectional energy transfers of the system **10**. In an embodiment, the EVSE **22** is a wall box that may be mounted to a wall **25** of the structure **14**. A charge cable **24** may operably connect the EVSE **22** to a charge port assembly **26** of the vehicle **12** for transferring energy between the vehicle **12** and the structure **14**. The charge cable **24** may be configured to provide any level of charging (e.g., 120 VAC, 240 VAC, Direct Current (DC) charging, etc.).

[0036] The EVSE **22** may be operably connected to an AC infrastructure **30** of the structure **14** through a home energy management system **28**. The home energy management system **28** may include a combiner box **40**, which is a type of bidirectional energy transfer module. Various electrical loads **31**, such as household appliance loads, for example, may be associated with the AC infrastructure **30**. The electrical loads **31** may sometimes be referred to as transient loads of the AC infrastructure **30** and could include loads associated with common kitchen appliances, washers, dryers, water heaters, air conditioning units, furnaces, home alarms systems, sump pump systems, routers, home lighting systems, etc. The AC infrastructure **30** may further include a main service panel **33** that is operably positioned between the combiner box **40** and the electrical loads **31**.

[0037] Power from a grid power source **32** (e.g., AC power), a renewable power source **34** (e.g., solar power, wind power, etc.), the vehicle **12**, or some combination of these may be selectively transferred to the AC infrastructure **30** for powering the electrical loads **31**. The combiner box **40**

can control power transfer to the AC infrastructure **30**. The combiner box **40** can in some implementations further control power transfer to the grid power source **32**. For example, power from the vehicle **12** could periodically be transferred through the combiner box **40** back to the grid power source **32**.

[0038] The combiner box **40** may function as a junction between the AC infrastructure **30** and each of the grid power source **32**, the vehicle **12**, and the renewable power source **34**. The combiner box **40** can electrically isolate the AC infrastructure **30** from the grid power source **32** during a grid power outage condition and can control the electrical loads **31** ON and OFF. In an embodiment, the combiner box **40** can activate a transfer switch to decouple the grid power source **32** from the electrical loads **31**, and then transfer power from the renewable power source **34**, the vehicle **12**, or both to the electrical loads **31**.

[0039] Power received from or transferred to the vehicle **12** may be transferred through the combiner box **40**. The combiner box **40** is configured to aid bidirectional transfers of electrical energy between the vehicle **12** and the structure **14**. The home energy management system **28** may include various equipment necessary for achieving the transfers of energy to/from the vehicle **12**.

[0040] The vehicle **12** may further include a vehicle power transfer system **36** configured for further enabling the bidirectional transfer of power between the vehicle **12** and the structure **14**. The vehicle power transfer system **36** may be operably connected between the charge port assembly **26** and the traction battery pack **16** of the vehicle **12**. The vehicle power transfer system **36** may include various equipment for enabling the vehicle **12** to act as a backup power source for transferring power to the structure **14**, such as a charger, a converter, an inverter, HV relays or contactors, a motor controller (which may be referred to as an inverter system controller or ISC), etc. The vehicle power transfer system **36** may further be configured to enable the vehicle **12** to receive power from the structure **14** and for transferring energy between the traction battery pack **16** and one or more electric motors of the vehicle **12**.

[0041] One non-limiting example of a suitable vehicle power transfer system that may be employed for use within the vehicle **12** for achieving bidirectional power transfers is disclosed within US Patent Publication No. 2020/0324665, assigned to Ford Global Technologies, LLC, the disclosure of which is incorporated herein by reference. However, other power transfer systems could also be utilized for achieving bidirectional power transfers within the scope of this disclosure.

[0042] FIG. **1** schematically illustrates a first configuration C1 of the system **10**. During the first configuration C1, power may be transferred from the structure **14** to the vehicle **12**, such as for charging the traction battery pack **16** of the vehicle **12**. The direction of energy transfer during the first configuration C1 is schematically depicted by arrow **38**.

[0043] FIG. **2** schematically illustrates a second configuration C2 of the system **10**. During the second configuration C2, power may be transferred from the traction battery pack **16** of the vehicle **12** to the structure **14**. The direction of energy transfer during the second configuration C2 is schematically illustrated by arrow **39**. In this way, the vehicle **12** may be employed as a backup energy management system for powering the electrical loads **31** of the structure **14**, such as when power from the grid power source **32** is temporarily unavailable as a result of electrical blackouts, for example.

[0044] The EVSE **22** may be configured to communicate with both the vehicle **12** and the combiner box **40** to facilitate transfers of power from the vehicle **12** to the structure **14** for powering the electrical loads **31** of the AC infrastructure **30**. The EVSE **22** is operably coupled to the vehicle **12** when transferring power from the vehicle **12**. The EVSE **22** is operably coupled to the structure **14** through the combiner box **40**.

[0045] As can be appreciated, the EVSE **22** and the combiner box **40** are ordinarily powered by the grid power source **32**. However, during a grid power outage condition where energy from the grid power source **32** is temporarily unavailable, the EVSE **22** and the combiner box **40** still need to be powered to be able to detect the vehicle **12** and to communicate with one another for transferring

power from the vehicle **12** to the structure **14**. This is sometimes referred to as a “dark start.”

[0046] The home energy management system **28** may additionally include a dark start energy storage resource **42** that is operably connected to the combiner box **40**. Although shown separately from the combiner box **40** in FIGS. **1-2**, the combiner box **40** and the dark start energy storage resource **42** could be integrated together as part of a common module (see, e.g., FIG. **3**).

[0047] The dark start energy storage resource **42** may be capable of supplying a limited amount of backup power to certain components of the system **10**, such as the EVSE **22**, for example, when power is unavailable from either the grid power source **32** or any other energy resource (e.g., the renewable power source **34**). The dark start energy storage resource **42** may include a battery (e.g., a 13V battery), a capacitor, a supercapacitor or any other suitable energy storage device.

[0048] Referring now primarily to FIG. **3**, the EVSE **22** may include internal circuitry **44**, a filter **46**, and a switched-mode power supply (SMPS) **48**. The circuitry **44** may include the components necessary for the EVSE **22** to function to transfer energy between the vehicle **12** and the structure **14**. The SMPS **48** may be a dual-mode SMPS that can receive input power from either the vehicle **12** or from the grid power source **32** through the combiner box **40**. During normal operating conditions of the grid power source **32**, power, such as 240 VAC power, for example, may be transferred to the filter **46** and then to the SMPS **48** from the grid power source **32**. The SMPS **48** may convert the 240 VAC input from the grid power source **32** into a 12 VDC output that can be used to power the circuitry **44** of EVSE **22**. The EVSE **22** and the combiner box **40** are therefore sufficiently powered to enable communications therebetween and with the vehicle **12** for preparing for and initiating the transfer of energy between the vehicle **12** and the structure **14**.

[0049] As further discussed below, the EVSE **22** and the combiner box **40** may each include features that enable the EVSE **22** and the combiner box **40** to maintain communications with one another and with the vehicle **12** even when the grid power source **32** is unable to supply power to the system **10**. These features, notably, do not require the EVSE **22** to include a battery that is sized to power every function of the EVSE **22**. Accordingly, in this embodiment, the EVSE **22** is considered batteryless, which can reduce the overall packaging requirements of the EVSE **22**.

[0050] The combiner box **40** may include a variable voltage regulated boost circuit **50** that is operably connected to the dark start energy storage resource **42**. The variable voltage regulated boost circuit **50** may be configured to selectively boost an output voltage received from the dark start energy storage resource **42**. Boosted power can then be transferred to the EVSE **22** for powering the circuitry **44** over a cable **64** that extends from the EVSE **22** to the combiner box **40**. The output voltage from the dark start energy storage resource **42** may need to be boosted before being transferred to the EVSE **22** in order to account for a voltage drop that can occur due to cable resistance across a length of the cable **64** when the distance between the EVSE **22** and the combiner box **40** is relatively large (e.g., greater than 100 feet).

[0051] In an embodiment, the cable **64** is a Power over Ethernet (POE) cable. Accordingly, both power and data can be sent to the EVSE **22** over the cable **64**.

[0052] The EVSE **22** may include a dark start measurement circuit **52**. The dark start measurement circuit **52** may be configured to measure the boosted output voltage received from the variable voltage regulated boost circuit **50** when the dark start energy storage resource **42** is supplying backup power during a grid power outage condition. As further explained below, the output voltage measured by the dark start measurement circuit **52** can be used to determine a voltage drop over the length of the cable, which can then be used to adjust the calibrated output voltage provided by the variable voltage regulated boost circuit **50**.

[0053] A control system **54** may control various functions associated with the system **10**, and in particular may be programmed to control the EVSE **22** and the combiner box **40** to maintain communications and facilitate power export from the vehicle **12** to the structure **14** during grid power outage conditions. The control system may include at least a first control module **56** and a second control module **58**. The first control module **56** may be a component of the combiner box

40, and the second control module **58** may be a component of the EVSE **22**. Although two control modules are specifically shown, the system **10** could include a greater number of controllers that are operably linked and configured to function together for facilitating various control strategies associated with the system **10**.

[0054] The first control module **56** and the second control module **58** may each include a processor **60** and non-transitory memory **62** for executing various control strategies and modes associated with the system **10**. The processor **60** can be a custom made or commercially available processor, a central processing unit (CPU), or generally any device for executing software instructions. The memory **62** can include a combination of volatile memory elements and nonvolatile memory elements. Volatile memory elements require power to store data, whereas nonvolatile memory elements can store data without consuming power. The processor **60** may be operably coupled to the memory **62** and may be configured to execute one or more programs stored in the memory **62** based on the various inputs received from other devices, such as the dark start energy storage resource **42**, the variable voltage regulated boost circuit **50**, the dark start measurement circuit **52**, etc.

[0055] FIG. **4**, with continued reference to FIGS. **1-3**, schematically illustrates in flow chart form an exemplary method **100** for controlling the system **10** during a grid power outage condition. In particular, the system **10** may be controlled for efficiently using dark start reserve power from the dark start energy storage resource **42** to maintain communications between the EVSE **22** and the combiner box **40** during the grid power outage condition. The system **10** may be configured to employ one or more algorithms adapted to execute at least a portion of the steps of the exemplary method **100**. For example, the method **100** may be stored as executable instructions in the memory(ies) of the control system **54**, and the executable instructions may be embodied within any computer readable medium that can be executed by the processor(s) of the control system **54**.

[0056] The exemplary method **100** may begin at block **102**. At block **104**, the method **100** may confirm whether or not power is available from the grid power source **32**. If NO, thus indicating a grid power outage condition, the method **100** may proceed to block **106** by determining whether or not power is available from the renewable power source **34** or any other source. If block **106** returns a NO flag, the method **100** may proceed to block **108**.

[0057] At block **108**, the combiner box **40** may send both a calibrated output voltage setting (data) and a first output voltage (power) from the dark start energy storage resource **42** to the EVSE **22**. The calibrated output voltage setting may be a preset value that is large enough to account for a worst case voltage drop over the cable **64** (e.g., 20V).

[0058] The dark start measurement circuit **52** of the EVSE **22** may measure the first output voltage at block **110** to determine a measured output voltage received by the EVSE **22**. The measured output voltage may be compared to the calibrated output voltage setting (such as via the second control module **58**, for example) to determine a voltage drop (data) between the EVSE **22** and the combiner box **40** at block **112**. In an embodiment, the voltage drop is the difference between the calibrated output voltage setting and the first output voltage. At block **114**, the voltage drop information may be sent to the combiner box **40** over the cable **64**.

[0059] Next, at block **116**, the variable voltage regulated boost circuit **50** (such as via commands from the first control module **56**, for example) may adjust the output voltage from the dark start energy storage resource **42** to a second output voltage based on the voltage drop. In an embodiment, the second output voltage is calculated by adding the voltage drop value to a minimum required voltage output that is necessary for maintaining proper control pilot operation between the EVSE **22** and the vehicle **12**. In an embodiment, the minimum required voltage output is 12V.

[0060] Since communications have been maintained between the EVSE **22** and the combiner box **40** using dark start energy, the method **100** may next proceed to block **118** and prepare the system **10** for transferring energy from the vehicle **12** to the structure **14**. The vehicle **12** can then transfer

energy to return power to the structure **14** at block **120**. The method **100** may end at block **122**.
[0061] The bidirectional energy transfer systems described herein include a dark start energy storage resource that can power system loads during grid power outage conditions. An output voltage of the dark start energy storage resource can be selectively boosted by a variable voltage regulated boost circuit to provide a calibrated output voltage to the EVSE that compensates for voltage drops. The proposed systems therefore eliminate the need for providing a dedicated power source within the EVSE and can further reduce the size of the dark start energy storage resource that is necessary to maintain operability of the system. Further, because the output of the boost circuit is a regulated voltage supply, the EVSE also does not require a secondary switching mode power supply, thereby further reducing components and complexity within the EVSE.
[0062] Although the different non-limiting embodiments are illustrated as having specific components or steps, the embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with features or components from any of the other non-limiting embodiments.
[0063] It should be understood that like reference numerals identify corresponding or similar elements throughout the several drawings. It should be understood that although a particular component arrangement is disclosed and illustrated in these exemplary embodiments, other arrangements could also benefit from the teachings of this disclosure.
[0064] The foregoing description shall be interpreted as illustrative and not in any limiting sense. A worker of ordinary skill in the art would understand that certain modifications could come within the scope of this disclosure. For these reasons, the following claims should be studied to determine the true scope and content of this disclosure.

Claims

1. A bidirectional energy transfer system, comprising: an electric vehicle supply equipment (EVSE); and a combiner box including a dark start energy storage resource and a variable voltage regulated boost circuit configured to boost an output voltage of the dark start energy storage resource for powering a circuitry of the EVSE during a grid power outage condition.
2. The bidirectional energy transfer system as recited in claim 1, wherein the dark start energy storage resource is a battery, a capacitor, or a supercapacitor.
3. The bidirectional energy transfer system as recited in claim 1, wherein the variable voltage regulated boost circuit is configured to boost the output voltage to compensate for a voltage drop that can occur across a length of a cable that extends from the EVSE to the combiner box.
4. The bidirectional energy transfer system as recited in claim 3, wherein the cable is a Power over Ethernet (POE) cable.
5. The bidirectional energy transfer system as recited in claim 1, comprising a control system programmed to control the variable voltage regulated boost circuit for boosting the output voltage during the grid power outage condition.
6. The bidirectional energy transfer system as recited in claim 5, wherein the control system includes a first control module of the combiner box and a second control module of the EVSE.
7. The bidirectional energy transfer system as recited in claim 6, wherein the second control module is programmed to determine a voltage drop based on a comparison between a measured output voltage received from the variable voltage regulated boost circuit and a calibrated output voltage setting received from the first control module.
8. The bidirectional energy transfer system as recited in claim 7, wherein the first control module is programmed to determine a calibrated output voltage for powering the circuitry based on the voltage drop.
9. The bidirectional energy transfer system as recited in claim 8, wherein the first control module is

programmed to command the variable voltage regulated boost circuit to adjust the output voltage to the calibrated output voltage for powering the circuitry.

10. The bidirectional energy transfer system as recited in claim 1, wherein the EVSE includes a dark start measurement circuit configured for measuring the output voltage.

11. The bidirectional energy transfer system as recited in claim 10, wherein the EVSE further includes a filter and a switched-mode power supply (SMPS).

12. A method, comprising: boosting an output voltage of a dark start energy storage resource of a combiner box during a grid power outage condition; and powering communications between an electric vehicle supply equipment (EVSE) and an electrified vehicle using the boosted output voltage.

13. The method as recited in claim 12, wherein the combiner box includes a variable voltage regulated boost circuit that is configured to boost the output voltage.

14. The method as recited in claim 12, wherein boosting the output voltage includes: sending a calibrated output voltage setting and a first output voltage from the combiner box to the EVSE.

15. The method as recited in claim 14, wherein boosting the output voltage further includes: measuring the first output voltage within a dark start measurement circuit of the EVSE to determine a measured output voltage received by the EVSE.

16. The method as recited in claim 15, wherein boosting the output voltage includes: comparing the measured output voltage to the calibrated output voltage setting; and determining a voltage drop.

17. The method as recited in claim 16, wherein boosting the output voltage includes: sending the voltage drop to the combiner box.

18. The method as recited in claim 17, wherein boosting the output voltage includes: adjusting the first output voltage to a second output voltage that is derived from the voltage drop.

19. The method as recited in claim 18, wherein the second output voltage is a sum of the voltage drop plus a minimum required voltage output that is necessary for maintaining a proper control pilot operation between the EVSE and the electrified vehicle.

20. The method as recited in claim 12, comprising, in response to establishing the communications between the EVSE and the electrified vehicle: transferring power from the electrified vehicle to a structure that is separate from the electrified vehicle during the grid power outage condition.
